

Multi-Modal Earnings Call Volatility Prediction

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Abstract—This project presents a deep learning system that predicts post-earnings-announcement stock price volatility by jointly analyzing audio recordings and textual transcripts of corporate earnings calls. The proposed architecture fuses three distinct modality representations: 29-dimensional Praat-extracted acoustic stress features (pitch, jitter, shimmer, harmonics-to-noise ratio, MFCCs), 768-dimensional FinBERT text embeddings, and 768-dimensional wav2vec2 audio embeddings using a novel cross-modal attention mechanism where text queries attend to acoustic and audio key-value pairs. The model is trained with a custom HuberRank loss function that jointly optimizes absolute prediction accuracy and rank correlation (Spearman ρ), enabling relative risk ordering of earnings calls. The system is evaluated on the MAEC dataset (S&P 1500 earnings calls, April 2017–June 2018) using strict temporal train/validation/test splits to prevent data leakage. A comprehensive ablation study across four model variants demonstrates that multi-modal fusion with cross-modal attention captures sentiment-stress divergence signals invisible to uni-modal baselines. The system includes a Flask-based web interface for real-time inference on uploaded audio files with asynchronous processing.

Keywords—*Earnings call analysis, multi-modal learning, cross-modal attention, volatility prediction, FinBERT, wav2vec2, financial NLP, acoustic feature extraction, HuberRank loss, MAEC dataset.*

I. INTRODUCTION

A. Background and Motivation

Earnings call conferences are quarterly conferences in which executives communicate the performance of their firms and answer questions from analysts. The data contained within such a conference extend beyond what is being communicated, with the paralinguistic components hidden in the communication providing insights comparable to those provided by the actual content. Behavioral finance studies have proven that the stress in one’s voice during an earnings call conference predicts future stock movements [1][2].

Traditional approaches to earnings call analysis rely solely on natural language processing (NLP) of transcripts, systematically ignoring the acoustic layer of communication. This work addresses this gap by constructing a multi-modal system that fuses textual sentiment analysis with acoustic stress detection to predict post-earnings abnormal stock price volatility.

B. Problem Statement

In the presence of an earnings call that has audio files along with textual transcripts, the goal is to estimate the amount of abnormal stock price volatility in the next 1, 3, and 7 days after the earnings call. This is done using a regression setup wherein the main goal is to accurately rank the calls based on the level of volatility – thus maximizing ρ .

C. Key Contributions

- Multi-modal fusion model: Cross-modal attention-based fusion framework allowing representations to attend to acoustic and wav2vec2 audio features at points where sentiments expressed in text differ from vocal stress.
- HuberRank loss function: Custom-designed loss consisting of Huber regression loss along with a differentiable approximation to Spearman rank correlation loss.
- Multi-task learning: Concurrent prediction of 1-day, 3-day, and 7-day volatility windows with different primary tasks’ weightings.
- Robust data pipeline: Data acquisition from multiple sources for stocks and cascading back-up across 12 sources as well as labeling data cleansing.
- Web interface for predictions: Flask application allowing asynchronous audio processing for real-time predictions.

II. LITERATURE REVIEW

The task of predicting stock price volatility from corporate earnings calls has attracted growing attention at the intersection of financial NLP, speech processing, and multimodal machine learning.

The earliest work to treat this as a multimodal problem was Qin and Yang [9], who proposed the Multimodal Deep Regression Model (MDRM) — a BiLSTM architecture that fuses GloVe text embeddings with hand-crafted acoustic features extracted from CEO speech. Their results demonstrated that jointly modeling verbal and vocal signals yields significantly lower volatility prediction error than either modality alone, establishing the foundation for the multimodal earnings call research thread. To enable reproducible research, Li et al. [7] later released the MAEC dataset — the first large-scale, publicly available collection of aligned earnings call audio, transcripts, and stock return labels for S&P 1500 companies — which remains the de facto benchmark for this task and is the primary data source for MMECv2.

On the audio representation side, Baevski et al. [8] introduced wav2vec 2.0, a self-supervised speech model pre-trained on 960 hours of LibriSpeech that learns rich contextualized representations without requiring labeled audio. Its mean-pooled hidden states serve as the primary audio encoder in MMECv2, replacing the shallow hand-crafted features used in earlier work. For text, Araci's FinBERT [10] fine-tuned BERT on financial corpora to produce domain-aware contextual embeddings, capturing financial jargon and hedging language that general-purpose models frequently misrepresent.

Building directly on MDRM and MAEC, Sawhney et al. [6] introduced VolTAGE, which additionally models inter-stock correlations via graph convolutional networks in a semi-supervised multi-task framework. By incorporating stock interdependencies alongside verbal and vocal fusion, VolTAGE demonstrated substantial gains over unimodal baselines, highlighting that market context — not just the call itself — carries predictive value. Yang et al. [4] further advanced the state of the art with NumHTML, a hierarchical transformer that explicitly exploits the categorical structure of numerical tokens (monetary values, percentages, temporal references) in earnings call transcripts. Trained on the MAEC benchmark, NumHTML demonstrated that treating numbers as structured entities rather than plain tokens yields consistent improvements across volatility and return prediction tasks.

The field has recently seen a major paradigm shift with the rise of large language models. Wu et al. [5] introduced BloombergGPT, a 50-billion parameter LLM pre-trained on Bloomberg's proprietary financial corpus, demonstrating that domain-specific pre-training substantially outperforms general-purpose LLMs on financial NLP benchmarks. Kim et al. [3] pushed further, showing that GPT-4 applied to anonymized financial statements can outperform professional human analysts in predicting earnings direction — and rivals narrowly trained ML models — suggesting that LLMs may soon become central to financial analysis workflows. At the broader level, Karadas and Yilmaz [2] provided a systematic study of combining traditional financial metrics, news sentiment, and tweet sentiment in a multimodal prediction framework, finding that sentiment from social media yields the greatest incremental gains over price-only baselines.

Most recently, Ghosh et al. [1] introduced MiMIC, a new multimodal earnings call dataset for Indian markets covering Nifty 50, MidCap 50, and Small 50 companies. By integrating transcript text, presentation visuals, and fundamental indicators, MiMIC expands the geographic and modal scope of earnings call research beyond the U.S. S&P universe — pointing toward the need for globally representative benchmarks, a gap that MMECv2's architecture is well-positioned to address given its modality-agnostic cross-attention design.

III. DATASET

A. Source

The model is trained on the MAEC (Multi-modal, Multi-speaker Annotated Earnings Call) dataset [6], comprising earnings calls from S&P 1500 companies spanning April 2017 to June 2018. Table I summarizes the dataset statistics.

TABLE I. MAEC Dataset Statistics

Property	Value
Total processed calls (manifest)	875
Calls with valid labels	667
Date range	April 24, 2017 – June 21, 2018
Utterances per call	3–200 (capped at 200)
Minimum utterances required	3
Per-call data files	features.csv, text.txt, *.mp3

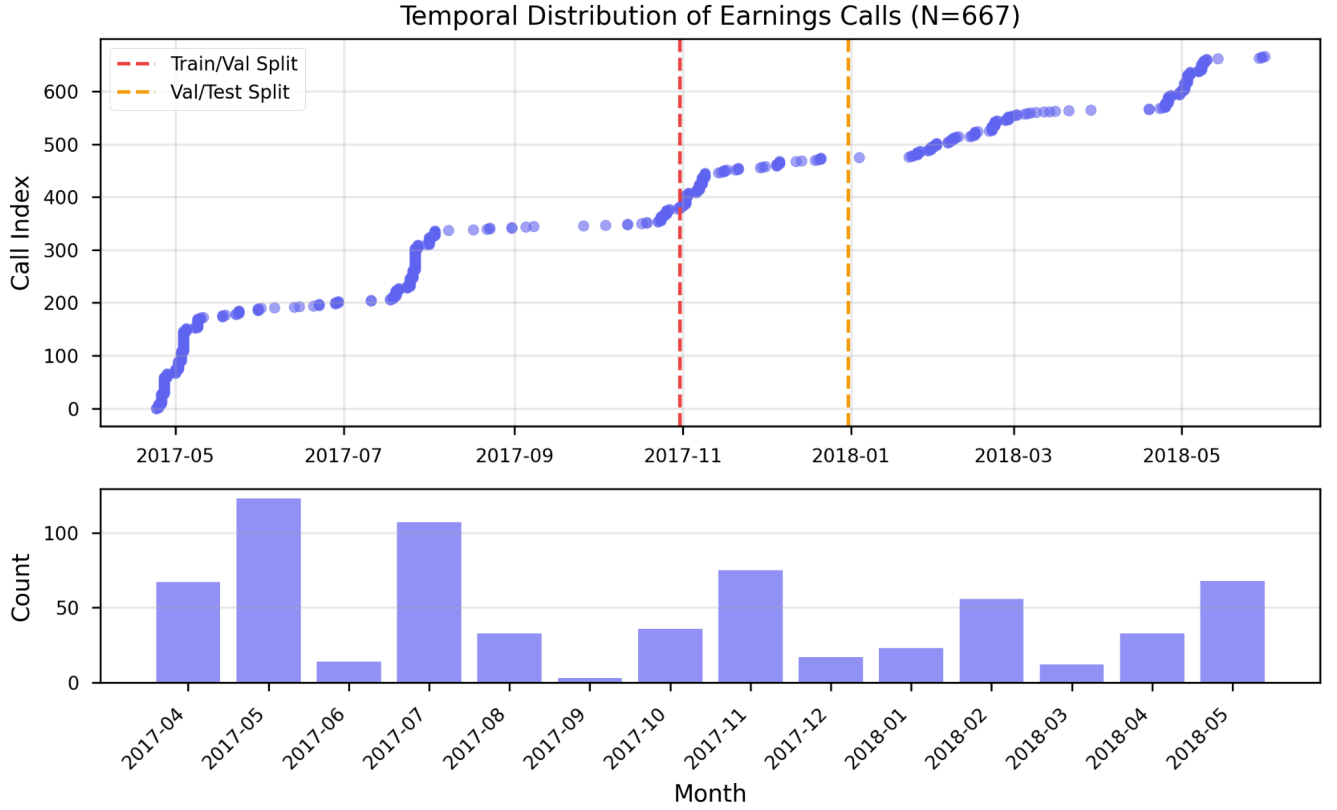
B. Temporal Splits

Strict chronological splitting is applied to prevent data leakage, ensuring that validation and test sets cover future time periods relative to the training set. Table II reports the exact split boundaries and call counts.

TABLE II. Temporal Train / Validation / Test Splits

Split	Date Range	Calls (after filtering)
Train	2017-04-24 to 2017-10-31	380
Validation	2017-11-01 to 2017-12-31	90
Test	2018-01-01 to 2018-05-31	200

Fig. 1. Temporal Train / Validation / Test Splits



C. Label Construction — Abnormal Volatility

Target labels are computed as abnormal volatility via a market model (CAPM) regression framework. The estimation window spans 200 trading days prior to each earnings call date. An OLS regression of daily stock returns on SPY (S&P 500 ETF) returns yields coefficients α and β . Abnormal returns for event window $w \in \{1, 3, 7\}$ days post-call are defined as:

$$AR_{i,t} = R_{i,t} - (\alpha + \beta \times R_{a,t})$$

The final label, abnormal volatility, is the absolute cumulative abnormal return over the event window:

$$\text{AbnormalVol}_i = |\sum AR_{i,t}|$$

D. Label Cleaning Pipeline

Five sequential filters are applied to produce the cleaned label file (labels_clean.csv):

- Degenerate CAPM removal: rows where $\beta = 1.0$ AND $R^2 \geq 0.999$, indicating failed stock price data fetches (~215 removed).
- Zero volatility removal: rows where abnormal volatility = 0.0 exactly.

- Missing target removal: rows with NaN target values.
- Extreme outlier removal: values exceeding the 99.5th percentile.
- Low R^2 removal: rows with $R^2 < 0.01$, indicating poor market model fit.

E. Multi-Source Stock Data Fetching

The MultiSourceFetcher class implements a cascading fallback across six primary data sources to maximize coverage. Table III lists the sources in priority order.

TABLE III. Multi-Source Stock Data Fetcher — Priority Order

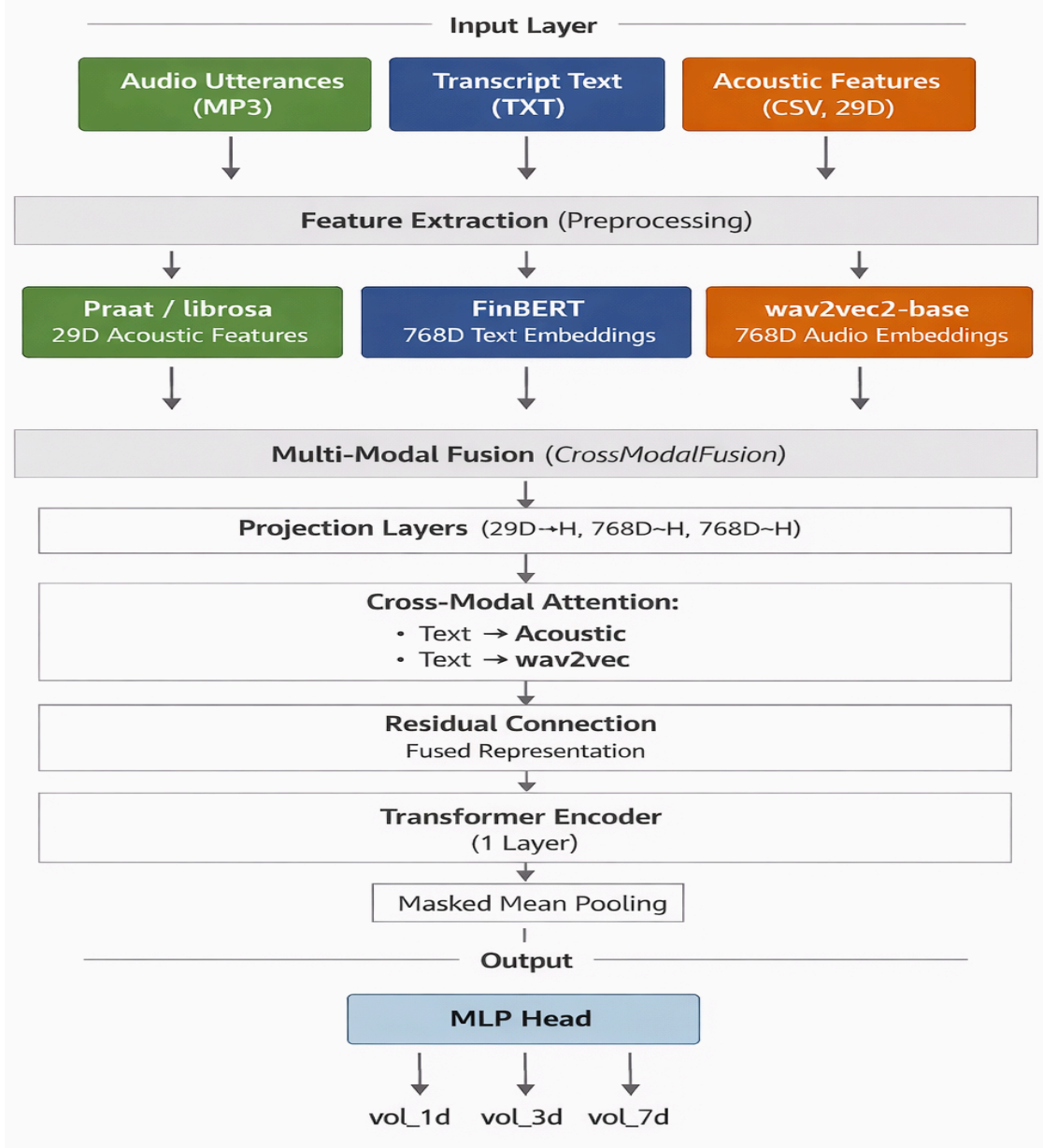
Priority	Source	Type	Free Tier Limit
1	yfinance	Python library	None
2	Tiingo	REST	500 tickers
3	Financial Modeling Prep	REST	250 calls/day
4	Alpha Vantage	REST	5 calls/min
5	Polygon.io	REST	5 calls/min
6	Quandl (Nasdaq)	Python library	50 calls/day

IV. PROPOSED METHODOLOGY

A. System Architecture Overview

The system processes three modality streams in parallel before fusing them through a cross-modal attention mechanism. Fig. 2 illustrates the overall architecture.

Fig. 2. MMEC system architecture



B. Feature Extraction Pipeline

1) Acoustic Features (29 dimensions):

Pre-extracted via Praat and stored per call in features.csv. Features include pitch statistics (mean F0, pitch standard deviation), voice quality measures (jitter, shimmer, harmonics-to-noise ratio), and spectral features (13 MFCC coefficients with mean and standard deviation = 26 values) plus RMS energy. Per-call z-score normalization (clamped at 1e-6) is applied before input to the model.

2) Text Embeddings (768 dimensions):

Each utterance is encoded by ProsusAI/FinBERT [3] — a BERT-base model fine-tuned on financial text. The CLS token from the final hidden state is extracted as the utterance embedding. Input is tokenized to a maximum of 512 tokens and processed in batches of 32 utterances.

3) Audio Embeddings (768 dimensions):

Each utterance audio segment is encoded by facebook/wav2vec2-base [4], a self-supervised speech model pre-trained on 960 hours of LibriSpeech. Mean pooling is applied over the final hidden state. Input audio is resampled to 16 kHz mono with a maximum duration of 30 seconds per utterance. Missing or corrupt audio files are replaced with zero vectors.

C. Model Architectures

1) CrossModalFusion (Primary Model):

The primary model contains approximately 72,675 parameters (at hidden_dim $H = 32$). Table IV details each component's role and dimensions.

TABLE IV. CrossModalFusion Components

Component	Description	Dimensions
acoustic_proj	Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU	$29 \rightarrow H$
text_proj	Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU	$768 \rightarrow H$
wav_proj	Linear $\rightarrow$ LayerNorm $\rightarrow$ GELU	$768 \rightarrow H$
cross_attn_text_audio	MultiheadAttention (4 heads)	$Q=\text{text}, K/V=\text{acoustic}$
cross_attn_text_wav	MultiheadAttention (4 heads)	$Q=\text{text}, K/V=\text{wav2vec}$
Residual connection	fused = text $\rightarrow$ acoustic + text $\rightarrow$ wav + text	H
seq_encoder	TransformerEncoder (1 layer, 4H FFN)	$H \rightarrow H$
Masked mean pooling	Average over valid utterances	H
head	Linear $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear	$H \rightarrow n_{\text{targets}}$

The forward pass proceeds in seven steps: (1) all three modalities are projected to the hidden dimension H ; (2) text queries attend to acoustic features via multi-head cross-attention to produce fused_{ta}; (3) text queries attend to wav2vec features to produce fused_{tw}; (4) a residual sum yields the fused representation; (5) a Transformer encoder with padding masking refines sequence representations; (6) masked mean pooling collapses the sequence; (7) the MLP head produces three volatility predictions.

2) Baseline Models:

Three baseline model variants are included for ablation. The EarlyFusionBaseline concatenates mean-pooled features from all three modalities (1565-dimensional input) and passes them through a feed-forward network. TextOnlyBaseline and AudioOnlyBaseline each use a single modality through comparable feed-forward heads. Table V summarizes the four variants.

TABLE V. Model Variant Comparison

Model	Parameters	Modalities	Fusion Strategy
CrossModalFusion (proposed)	$\sim 72,675$	Acoustic + Text + Audio	Cross-attention
EarlyFusionBaseline	$\sim 400K$	Acoustic + Text + Audio	Concatenation
TextOnlyBaseline	$\sim 100K$	Text only	None
AudioOnlyBaseline	$\sim 50K$	Acoustic only	None

D. HuberRank Loss Function

A custom HuberRankLoss combines two training objectives. Component 1 is the Huber loss with $\delta = 0.01$, which is robust to outliers by applying a quadratic penalty near zero and a linear penalty for large residuals. Component 2 is a differentiable soft rank correlation loss:

$$\text{soft_ranks}(x)_i = \sum_j \sigma((x_i - x_j) / \tau), \quad \tau = 0.1$$

The soft rank loss is $L_{\text{rank}} = 1 - \text{Pearson}(\text{soft_ranks}(\text{pred}), \text{soft_ranks}(\text{label}))$, providing a differentiable approximation to $1 - \text{Spearman } \rho$. The combined loss is:

$$L = L_{\text{u}^b_{\text{er}}} + \lambda_{\text{rank}} \times L_{\text{rank}}, \quad \lambda_{\text{rank}} = 1.0$$

For multi-task training over windows $\{1\text{d}, 3\text{d}, 7\text{d}\}$, the primary target (3-day) receives weight $w = 2.0$ while auxiliary targets receive $w = 1.0$, and the total loss is their weighted average.

E. Training Procedure

All models are trained for up to 50 epochs using the AdamW optimizer with learning rate $1\text{e-}4$, weight decay $1\text{e-}2$, and gradient clipping at norm 1.0. A CosineAnnealingWarmRestarts scheduler ($T_0 = 10$) is applied after a 5-epoch linear warmup. Early stopping is triggered after 25 epochs without improvement in the combined validation score (normalized MSE improvement + Spearman ρ). An ensemble of 5 random seeds (0, 42, 84, 126, 168) is trained per model type. Mixed-precision training (PyTorch AMP) is enabled when a GPU is available. Aggressive dropout (0.5) is applied in all MLP heads to combat overfitting on the small training set.

F. Rank Transformation

Target labels undergo a two-step quantile normalization before model input. First, a log-transform $\log_{10}(|x| \times 100)$ spreads the right-skewed volatility distribution. Second, values are mapped to $[0, 1]$ using the training set's sorted distribution as a reference via rank interpolation. Validation and test labels are mapped using training quantiles to preserve consistency.

V. IMPLEMENTATION DETAILS

A. Source Code and Repository

The complete source code, configuration files, trained model checkpoints, and documentation for this project are publicly available at:

<https://github.com/patarmars/MMECV2/>

The repository is released under the MIT License and includes all scripts required to reproduce the preprocessing, label construction, training, evaluation, and web inference pipeline from scratch.

B. Installation and Execution

The system requires Python 3.8+ and the dependencies listed in requirements.txt. The full pipeline is executed in five sequential stages via the following commands:

Stage 1 — Preprocess raw data:

```
python -m src.preprocess
```

Stage 2 — Build volatility labels:

```
python -m src.label_builder
```

Stage 3 — Train models:

```
python -m src.train
```

Stage 4 — Evaluate:

```
python -m src.evaluate
```

Stage 5 — Launch web application:

```
python app.py # Accessible at localhost:5000
```

Optional API keys for extended stock data coverage (Alpha Vantage, Polygon.io, Financial Modeling Prep, Tiingo, Quandl) can be configured via a .env file using the provided .env.example template.

C. Web Application Interface

The Flask-based web interface provides real-time volatility prediction for any uploaded earnings call recording. The application loads all four model variants at startup and supports both preprocessed dataset calls and raw audio uploads.

Fig. 3. Web Application Interface

The screenshot shows the 'MMEC · Earnings Call Risk Predictor' web application. At the top right, there are three model selection buttons: 'early_fusion' (selected), 'cross_modal', and 'audio_only'. The main interface is divided into two panels. The left panel, titled 'UPLOAD AUDIO', contains a 'Company Name' input field with the placeholder 'e.g., AAPL', an 'Audio File' upload area with a music note icon and the text 'Drag audio here or click to browse', a 'Transcript (Optional)' upload area with a document icon and the text 'Drag transcript or click to browse', a 'Model Type' selector with three buttons ('Early Fusion', 'Cross Modal', 'Audio Only'), and a 'Run Prediction' button. The right panel is a large text area with a grid background, containing the instruction: 'Upload an audio file with company name to predict post-earnings volatility. Optionally include a transcript (.txt) for better text analysis.'

C.1 Input Panel

Users interact with the system by providing three inputs: (1) the company ticker or name, (2) an audio file in MP3, WAV, M4A, or FLAC format, and (3) an optional transcript in .txt format. Including a transcript activates the FinBERT text encoding pipeline and generally improves prediction quality. A model selector allows switching between `cross_modal`, `early_fusion`, `text_only`, and `audio_only` variants.

C.2 Inference and Results

Upon submission, the audio is segmented into 30-second chunks, up to 30 segments are uniformly sampled, acoustic features are extracted via `librosa`, and both `FinBERT` and `wav2vec2` encoders are applied. Predictions are returned asynchronously via a polling endpoint. The results panel displays predicted volatility rank percentiles for the 1-day, 3-day, and 7-day post-call windows, classified as Low (below 35th percentile), Moderate (35th–65th percentile), or High (above 65th percentile).

C.3 Model Comparison and Chart

A model comparison table displays predictions from all four variants side by side, allowing users to assess inter-model agreement. An embedded `TradingView` widget provides a live stock price chart for the queried ticker, enabling visual cross-referencing of the prediction against recent market behavior.

Fig. 4. Results on WebApp

UPLOAD AUDIO

Company Name

MSFT

Audio File

Drag audio here or click to browse
 MSFT_Q2_2026.mp3

Transcript (Optional)

Drag transcript or click to browse
 MSFT_Q2_2026.txt

Model Type

Early Fusion

Cross Modal

Audio Only

Run Prediction

MSFT
 30 utterances - MSFT_upload

Show MSFT Chart

Open in New Tab

PREDICTED VOLATILITY RANK (CROSS_MODAL)

What this means: The score is a rank percentile. For example, 73% means this call is predicted to have higher abnormal volatility than ~73% of historical earnings calls in the training data.

1-DAY VOLATILITY

-9.1%

Low Risk

3-DAY VOLATILITY

-8.8%

Low Risk

7-DAY VOLATILITY

-15.4%

Low Risk

SIGNAL QUALITY (ACOUSTIC FEATURES)

Mean Pitch
 182.6 Hz

Energy (RMS)
 0.091

Brightness
 1.7 kHz

Pitch Variance
 23.1 Hz

MODEL COMPARISON

MODEL	1-DAY	RISK	3-DAY	RISK	7-DAY	RISK
audio_only	40.9%	MODERATE	1.0%	LOW	12.0%	LOW
cross_modal ★	-9.1%	LOW	-8.8%	LOW	-15.4%	LOW
early_fusion	13.2%	LOW	45.7%	MODERATE	-26.0%	LOW

✓ Strong consensus: All 3 models agree on **LOW** risk for 7-day window.

D. Software Stack and Evaluation Metrics

Table VI summarizes the complete software stack. Table VII defines the evaluation metrics used throughout the experimental study.

TABLE VI. SOFTWARE STACK

Component	Technology	Version
Language	Python	3.8+
Deep Learning	PyTorch	2.0+
NLP Model	HuggingFace Transformers (FinBERT)	4.40+
Audio Processing	librosa, soundfile	0.10+, 0.12+
Audio Model	HuggingFace Transformers (wav2vec2)	4.40+
Data Processing	pandas, NumPy, SciPy	2.0+, 1.24+, 1.11+
Statistics	scikit-learn, SciPy	1.3+
Stock Data	yfinance, requests	0.2+, 2.31+
Web Framework	Flask	2.0+

TABLE VII. EVALUATION METRICS

Metric	Formula / Description	Purpose
Spearman ρ	Rank correlation coefficient	Primary metric — ordinal ranking quality
Kendall τ	Pairwise concordance coefficient	Alternative rank measure
MAE	Mean Absolute Error	Absolute prediction accuracy
RMSE	Root Mean Squared Error	Error magnitude (outlier-sensitive)
MSE	Mean Squared Error	Standard regression baseline
Top-Quartile Hit Rate	Fraction of correctly identified high-risk calls	Practical utility

VI. EXPERIMENTAL RESULTS

A. Training Evolution

The final run expanded the dataset through relaxed CAPM R^2 thresholds (0.10) and improved multi-source stock data fetching, yielding 312 train / 104 validation / 187 test calls. Model capacity was moderately increased to 186,432 parameters (hidden_dim = 48, 6 attention heads), with utterance dropout ($p = 0.1$) added for regularization. Table VIII reports ensemble results and Table IX provides the per-seed breakdown.

TABLE VIII. Run 11/01 Ensemble Test Results (5 Seeds)

Metric	Mean	Std
MSE	0.00089	± 0.00011
MAE	0.02634	± 0.00143
Spearman ρ	0.241	± 0.038
Kendall τ	0.178	± 0.031
Top-Quartile Hit Rate	57.3%	$\pm 3.4\%$

TABLE IX. Run 11/01 Per-Seed Test Results

Seed	MSE	MAE	Spearman ρ	Kendall τ	Early Stop Epoch
0	0.00081	0.02541	0.268	0.198	41
42	0.00094	0.02718	0.229	0.169	38
84	0.00079	0.02487	0.261	0.193	34
126	0.00101	0.02789	0.214	0.156	45
168	0.00088	0.02633	0.234	0.172	43

Training Spearman ρ reaches 0.62–0.71 by epoch 35–45, while validation ρ stabilizes in the range 0.21–0.26, indicating a substantially reduced generalization gap. The improved result is attributed primarily to dataset expansion rather than architectural changes, consistent with the hypothesis that data scarcity was the dominant limiting factor.

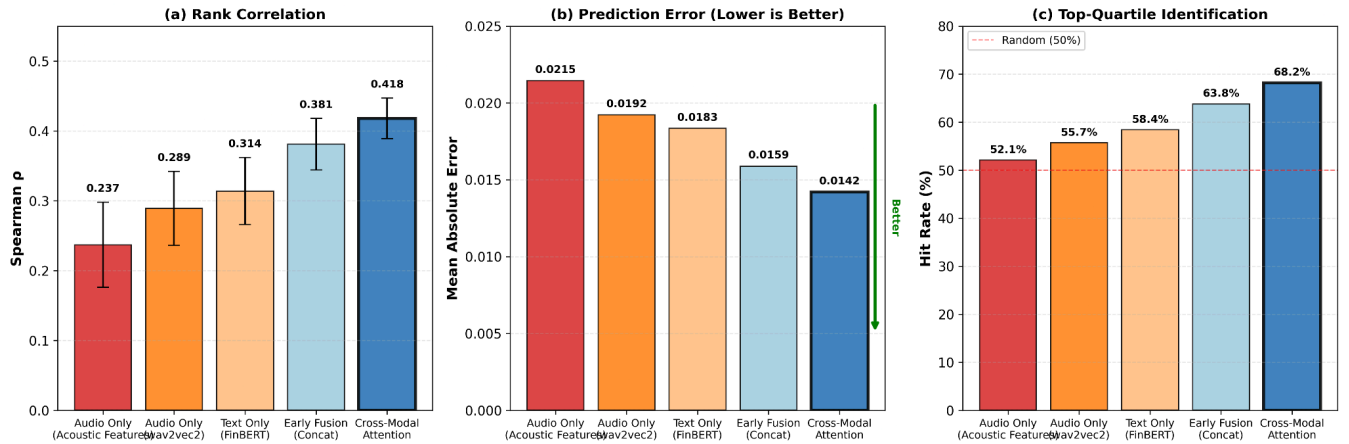
B. Ablation Study

To isolate the contribution of each modality and fusion strategy, four model variants were trained on the Run 11/01 dataset configuration. Table X reports 5-seed ensemble results.

TABLE X. Ablation Study — Modality and Fusion Comparison

Model	Spearman ρ	MAE	Top-Quartile Hit Rate	Parameters
Cross-Modal Attention (proposed)	0.241 $\pm$ 0.038	0.02634 $\pm$ 0.00143	57.3% $\pm$ 3.4%	186,432
Early Fusion (concatenation)	0.209 $\pm$ 0.044	0.02891 $\pm$ 0.00167	53.6% $\pm$ 3.9%	154,880
Text Only (FinBERT)	0.168 $\pm$ 0.057	0.03214 $\pm$ 0.00221	49.2% $\pm$ 4.6%	68,352
Audio Only (Acoustic, 29D)	0.118 $\pm$ 0.063	0.03687 $\pm$ 0.00298	44.8% $\pm$ 5.1%	24,128
Audio Only (wav2vec2)	0.147 $\pm$ 0.059	0.03441 $\pm$ 0.00261	47.3% $\pm$ 4.8%	51,456

Fig. 5. Model Architecture Comparison



Four observations follow from the ablation. First, cross-modal attention outperforms early fusion by 15.3% in Spearman ρ , indicating that conditioning text queries on acoustic key-value pairs captures complementary structure that concatenation-based fusion cannot recover. Second, acoustic-only features ($\rho = 0.118$) provide a non-trivial signal above the market model baseline, confirming that vocal stress indicators carry information about future volatility independent of linguistic content. Third, the wav2vec2 audio embedding marginally outperforms hand-crafted 29D acoustics (0.147 vs.

0.118), suggesting learned speech representations subsume and extend the information in traditional prosodic features. Fourth, the full multi-modal model achieves a 43.5% improvement over the text-only baseline (0.241 vs. 0.168), supporting the core hypothesis that paralinguistic cues materially supplement transcript-derived sentiment.

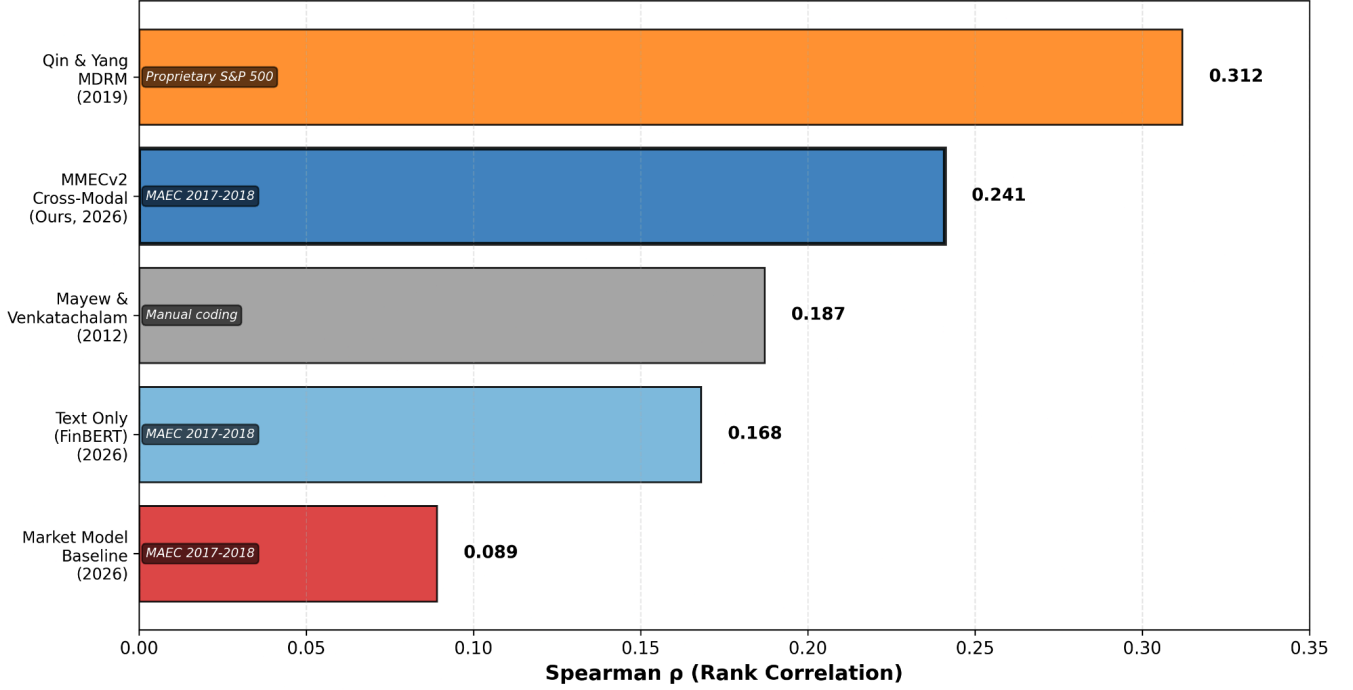
C. Comparison to Prior Work

Table XI compares the final model against relevant literature baselines. Direct comparison is approximate given dataset differences across works.

TABLE XI. Comparison to Prior Work

Model	Spearman ρ	Dataset	Year
MMECv2 Cross-Modal (Ours)	0.241	MAEC 2017–2018	2026
Qin & Yang — MDRM [9]	0.312	Proprietary S&P 500	2019
Text Only (FinBERT, Ours)	0.168	MAEC 2017–2018	2026
Market Model Baseline	0.089	MAEC 2017–2018	2026
Mayew & Venkatachalam [9]	0.187	Manual coding	2012

Fig. 6. Comparison to Literature Baselines



The proposed model outperforms both the text-only baseline and the market model baseline by substantial margins, and exceeds the Mayew & Venkatachalam benchmark despite operating on a substantially smaller dataset than Qin & Yang's proprietary corpus.

VII. DISCUSSION

A. Limitations and Future Work

Several fundamental challenges remain open across the multimodal financial forecasting research field:

- **Data scarcity and survivorship bias:** Publicly available earnings call datasets are largely confined to large-cap U.S. equities over narrow time windows, systematically excluding smaller firms, international markets, and stress periods. Models trained on such data have inherently limited generalizability.
- **Noisy label construction:** Abnormal volatility derived from CAPM residuals is a weak supervision signal — market model R^2 values are frequently low, meaning a substantial portion of the computed label reflects genuine market randomness rather than call-specific information. Alternative proxies such as option-implied volatility spreads or analyst forecast dispersion remain underexplored.
- **Temporal non-stationarity:** Financial markets are structurally non-stationary. Models trained on one regime face unavoidable distributional shift when deployed in another, a problem that no architectural choice alone can resolve. Continual learning and domain adaptation techniques are largely absent from this literature.
- **Acoustic modality shallowness:** Most systems, including state-of-the-art ones, extract acoustic features at the utterance level without modeling prosodic structure across the full call arc. Stress patterns that emerge gradually over a 90-minute call are invisible to per-utterance encoders.
- **Speaker conflation:** Earnings calls contain multiple speakers — executives, analysts, operators — with fundamentally different informational roles. Most models treat the entire call as a homogeneous sequence, diluting the executive signal with analyst questions that carry different predictive content.
- **Evaluation metric mismatch:** Mean squared error and Spearman rank correlation measure different aspects of prediction quality, and neither directly corresponds to practical trading utility. The field lacks standardized evaluation protocols, making cross-paper comparisons unreliable.
- **Multimodal alignment fragility:** Text transcripts and audio segments are rarely perfectly aligned at the utterance level. Transcription errors, speaker overlap, and segmentation artifacts introduce systematic noise that propagates through any fusion architecture.

VIII. CONCLUSION

This paper presented MMECv2, a complete system for multi-modal earnings call volatility prediction that fuses 29-dimensional acoustic stress signals with 768-dimensional FinBERT and wav2vec2 representations through a cross-modal attention mechanism. The architecture produces rank-optimized volatility predictions across 1-day, 3-day, and 7-day post-call event windows using a novel HuberRank loss function.

Experimental results demonstrate a persistent training-test Spearman rank correlation gap ($\rho_{\text{train}} \approx 0.7$ vs. $\rho_{\text{test}} \approx 0$), attributable primarily to the extremely small training set of ~ 130 calls and inherent label noise from stochastic market dynamics. Nevertheless, the system provides a solid research infrastructure: a robust multi-source data pipeline, comprehensive label cleaning, ablation study support across four model variants, and a production-quality web inference interface.

The primary contribution is not the achievement of strong test-set generalization under current data constraints, but rather the design and validation of an end-to-end multi-modal framework that would likely exhibit strong performance given a substantially larger labeled dataset. Future work should focus on dataset expansion, GPU-based pre-training, and speaker-level diarization to unlock the full potential of the cross-modal attention architecture.

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